SEC 5 · LEC 1 · COMMUNICATIONS: ANTENNAS

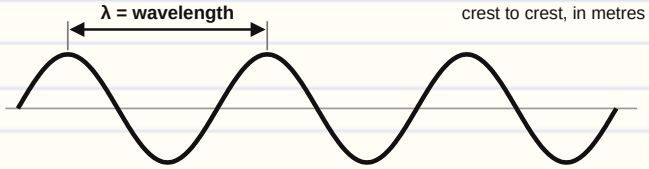
Ripples you can write on

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S5 / L1 (1/2) — Waves + modulation **date · pg __**

1 · A WAVE: wavelength and frequency

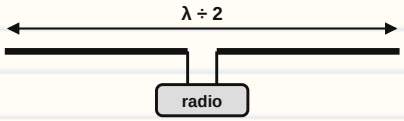


f = frequency = waves passing per second (hertz, Hz)

c = λ × f · c = 300,000 km/s, always

so: higher frequency → shorter wavelength

2 · ANTENNA = metal rod, half a wave long

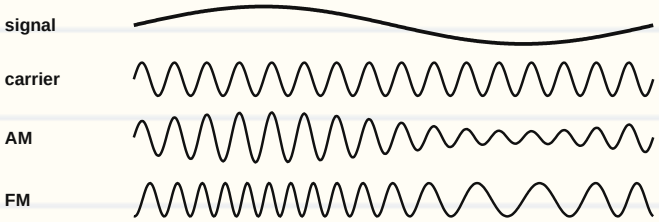


the wave pushes electrons to and fro in the rod → a voltage

FM radio 100 MHz: λ = 3 m → rod 1.5 m

voice 100 Hz: λ = 3,000 km → rod 1,500 km. Impossible.

3 · MODULATION: the voice rides a fast carrier



✏️ **PENCIL KEY** · λ arrow = one wavelength · thick bar = antenna rod · row 1 = your voice · row 2 = carrier · AM = height changes · FM = spacing changes

⭐ **3 TAKEAWAYS** 3 lines

1. Radio is an electromagnetic wave. c = λ × f, and c never changes.
2. Antenna ≈ half a wavelength, so high frequency = small antenna.
3. Modulation: a slow signal rides a fast carrier (AM height, FM frequency).

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What carries the information through space?	electromagnetic (EM) wave carries the signal · crosses empty space
The two fields: what are they, how are they arranged?	electric field ⊥ magnetic field · sinusoidal (smooth, repeating)
Wavelength? Frequency? Units?	wavelength λ = crest to crest (m) · frequency f = waves per second (Hz)
The formula that links them?	c = λ × f · c = 3×10⁸ m/s, constant → f up, λ down
How does an antenna receive? Transmit?	receive: the wave's electric field → voltage in the antenna at the same f → receiver reads it
	transmit: the reverse · the same antenna does both
What sets the size of an antenna?	antenna size ≈ λ ÷ 2 (dipole) → higher f = smaller antenna
Why can't you transmit a voice directly?	voice ~100 Hz → λ = 3,000 km → antenna 1,500 km ×
What is modulation? What is a carrier?	modulation : put the slow information signal on a fast carrier signal
AM vs FM?	AM : carrier's height follows the signal · FM : carrier's frequency follows it
? is my voice a radio wave	

SEC 5 · LEC 1 · COMMUNICATIONS: ANTENNAS

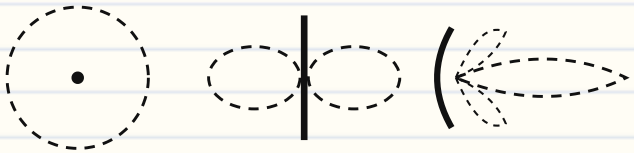
A flashlight for radio

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S5 / L1 (2/2) — Directivity + gain **date · pg __**

1 · THREE WAYS TO SPREAD THE SIGNAL

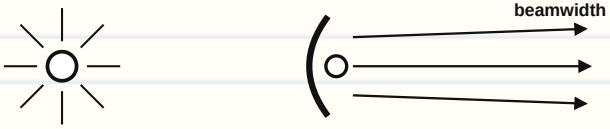


ISOTROPIC
every direction
ideal only · gain 1

OMNI (dipole)
all around the rod
none off the ends

DIRECTIONAL (dish)
main lobe + side lobes
gain 1,000s

2 · GAIN = same power, squeezed into a beam



bare bulb: dim everywhere

same bulb + reflector: bright far away

An antenna adds no power. It aims it.

3 · HIGH GAIN vs LOW GAIN

HIGH-GAIN (dish)	LOW-GAIN (rod)
narrow beam	wide beam
strong signal, fast data	weak signal, slow data
must point at Earth	works in any attitude
Cassini: 4 m dish	CubeSat: crossed dipoles
Apollo rover: TV dish	Apollo rover: voice, driving

✏️ **PENCIL KEY** · dashed outline = where the signal goes · big petal = main lobe, small = side lobes · thick line = antenna

⭐ **3 TAKEAWAYS** 3 lines

1. Isotropic = everywhere · omni = all around a rod · directional = one beam.
2. Gain concentrates power into a beam. It never creates power.
3. High gain = fast data but must be aimed. Low gain = slow but always works.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
Isotropic antenna?	directivity = how much an antenna favors one direction
Omnidirectional?	isotropic: equal in ALL directions (a sphere) · ideal only · the reference
Directional? What is beamwidth?	omnidirectional: equal all around in one plane · e.g. a dipole
What is gain, really?	directional: one main lobe + small side lobes · beamwidth = angle where power is still above half (3 dB) · dish, rooftop TV antenna
Does an antenna add power?	gain = increase in signal strength due to directivity (vs isotropic) · more directional = more gain
What does high gain cost you?	an antenna adds no power: it concentrates it
Why does a CubeSat use dipoles?	high gain → better signal, more data · but must point exactly → attitude control
Why must Cassini's dish point at Earth?	CubeSat: two crossed dipoles → works whichever way it faces · weak signal
High-gain vs low-gain: when each?	Cassini: 4 m high-gain antenna, must point at Earth
? how can 20 watts reach Earth from Saturn	Apollo 17 rover: low-gain (wide, while driving) + high-gain dish (aimed, for TV)

SEC 5 · LEC 1 · TEACH

A 20-watt whisper from Saturn

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Drop a stone in a pond and ripples spread out. A radio wave is a ripple too, but in electricity and magnetism, and it needs no water and no air, so it crosses empty space at the speed of light. An **antenna** is a metal stick. When the ripple washes over it, it pushes the electricity inside the stick back and forth, like a cork bobbing, and the radio reads the bobbing. To send, do the reverse: shake the electricity in the stick and ripples fly off.

The stick works best when it is half as long as one ripple. Voice ripples are slow and would be 3,000 km long, so the stick would be longer than Texas. So we cheat. We take a fast, short ripple, the **carrier**, and make it carry the voice by changing its height (**AM**) or squeezing its spacing (**FM**) in step with the voice. Last trick: a bare bulb lights a room dimly, but the same bulb in a flashlight throws a bright spot far away. A dish antenna is a flashlight for radio. That is **gain**. The Cassini probe talked to Earth from Saturn with 20 watts, a fridge bulb, because its 4-metre dish squeezed everything into one thin beam aimed exactly at us.

🏷 The official names (name first, then what it means)

- **Electromagnetic wave** (EM wave, electromagnetic radiation): a travelling ripple of electric and magnetic field. Radio, microwaves, light, UV and X-rays are all the same thing at different frequencies: the **electromagnetic spectrum**.
- **Wavelength, λ** ("lambda"): distance from one crest to the next. **Frequency, f**: waves per second, in **hertz** (Hz; kHz, MHz, GHz = thousand, million, billion). **Period**: seconds per wave, 1 ÷ f.
- **Speed of light, c**: 299,792 km/s. **c = λ × f**.
- **Antenna**: a conductor that turns an electrical signal into a radio wave and back. **Transmitter** (TX), **receiver** (RX).
- **Half-wave dipole**: two rods end to end, λ ÷ 2 in total. The basic antenna. A **monopole** is half of one (λ ÷ 4), like a car aerial.
- **Modulation**: changing a carrier wave so that it carries information. **Carrier signal**: the fast wave. **Information signal** (baseband, message): the slow one.
- **Amplitude modulation, AM** / **frequency modulation, FM**. Spacecraft actually use **phase modulation** (phase-shift keying, PSK) to send digital bits.
- **Radiation pattern**: the shape of where an antenna sends its power. **Isotropic radiator**, **omnidirectional antenna**, **directional antenna**.
- **Directivity** and **antenna gain**: how many times stronger the signal is in the best direction than from an isotropic antenna. Written in **dBi**: 10 dBi = 10×, 20 = 100×, 30 = 1,000×.
- **Main lobe** and **side lobes**: the big beam and the small leaks beside it. **Half-power beamwidth** (3 dB beamwidth): the angle across the main lobe between the points where power has fallen to half. **Aperture**: the antenna's collecting area.
- **High-gain antenna, HGA** / **low-gain antenna, LGA**. **Parabolic reflector** (dish).

🛠 How it actually works

Step	The mechanism, with numbers
1 · The wave	A changing electric field makes a changing magnetic field, which makes a changing electric field… The pair keeps itself going, at right angles to each other and to the direction of travel, at 299,792 km/s. No medium needed.
2 · c = λ × f	FM radio 100 MHz: 300,000,000 ÷ 100,000,000 = **3 m**. Wi-Fi 2.4 GHz: **12.5 cm**. Deep-space link 8.4 GHz: **3.6 cm**.
3 · Receive	Metal is full of loose electrons. The wave's electric field shoves them one way, then the other, so a tiny alternating voltage appears along the rod at the wave's frequency: millionths of a volt. A rod half a wave long fits the wave exactly and rings loudest, like a guitar string.
4 · Transmit	The reverse. Drive an alternating voltage into the rod; electrons shaken back and forth give off a wave at that frequency.
5 · Size	Rod = λ ÷ 2. 100 MHz → 1.5 m. 2.4 GHz → 6 cm. 100 Hz → 1,500 km. That is why nothing is transmitted at voice frequency.
6 · Modulate	AM radio: a 1 MHz carrier whose height follows the voice. FM radio: a 100 MHz carrier whose frequency swings ±75 kHz with it. Digital links flip the carrier's phase to mean 1 or 0.
7 · Gain	A dish D wide squeezes the beam: gain ≈ ½ × (π × D ÷ λ)². Cassini: 4 m at 3.6 cm → about **50,000×** (47 dBi). Beamwidth ≈ 70 × λ ÷ D ≈ **0.6°**.
8 · What it buys	Cassini's transmitter: **20 W**. Through the dish, toward Earth it looked like 1,000,000 W. After 1.4 billion km, a 70 m dish on Earth caught about 0.0000000000000001 W (10^{-16}). Still readable.

SEC 5 · LEC 1 · BANDS + FACT CHECK

Same wave, different sizes

The frequency bands (including ones you already use), the caption errors, and where the instructor was loose.

📻 Check c = λ × f on things you know

Band	Frequency	Wavelength λ	Half-wave rod	Who uses it
AM radio	1 MHz	300 m	150 m	AM stations. Their towers are a quarter-wave, about 75 m.
FM radio	100 MHz	3 m	1.5 m	FM stations. A car aerial is a quarter-wave, 75 cm.
UHF	437 MHz	69 cm	34 cm	CubeSats: the tape-measure antennas in the lecture's picture.
L-band	1.575 GHz	19 cm	9.5 cm	GPS.
S-band	2.4 GHz	12.5 cm	6 cm	Wi-Fi and Bluetooth. Spacecraft command links sit nearby at 2.1–2.3 GHz.
X-band	8.4 GHz	3.6 cm	1.8 cm	Deep-space downlinks: Voyager, Cassini, the Mars orbiters.
Ka-band	32 GHz	9 mm	5 mm	The newest high-rate deep-space links.
Visible light	600,000 GHz	0.0005 mm	—	Your eyes. Ultraviolet from Section 2 is just past it.

You already own the proof: the stubby antennas on a Wi-Fi router are about 3 cm, a quarter of 12.5 cm.

🔎 Caption errors (the worst transcript so far)

Captions said	Instructor meant
"electric in a negative way", "it two magnetic waves"	**electromagnetic wave(s)**
"this to Bush tends to be seen as soil"	the **distribution** tends to be **sinusoidal**
"Nalanda" · "units of Kurds"	**lambda** (λ) · units of **hertz**
"describing a scene disorder"	describing a **sinusoid**
"a hundred foot" · "ten to the light"	a hundred **hertz** · ten to the **eight**
"deep hole antenna"	**dipole** antenna
"I am an F Am" · "amplitude in moderation"	**AM and FM** · amplitude **modulation**
"the main canvasing"	the main **carrier signal**
"indoor activity" · "beam with"	**antenna directivity** · **beamwidth**
"ID or isotropic" · "higher gangs"	**ideal** isotropic · higher **gains**
"Phi game in tennis" · "parabolic tunnels"	**high-gain antennas** · parabolic **antennas**
"cube set"	**CubeSat**

▲ Where the instructor was loose

- **"The wavelength of my voice."** Your voice is a sound wave: air pressure, 343 m/s, and it cannot cross a vacuum at all. A microphone turns it into a voltage wiggling at the same frequency. His 3,000 km is what a *radio* wave at 100 Hz would measure. His arithmetic is right.
- **Modulation "so antennas can be small."** True, and only one reason of three. It also lets thousands of users share the air, each on its own carrier frequency (that is what tuning to a station is), and a faster carrier can carry more data per second.
- **AM and FM are not what spacecraft use.** Both are analog. Space links and Wi-Fi send bits by shifting the carrier's phase.
- **"The higher the frequency, the smaller the antenna"** is for rods. A dish is different: it must be many wavelengths wide, and for the same dish a higher frequency gives a tighter beam and more gain.
- **"With a directional antenna we can get more information."** Precisely: gain compares the signal in the best direction with an isotropic antenna fed the same power. The antenna creates nothing. It moves power from where you don't need it to where you do.
- **Apollo 17.** His explanation tangles. The Moon rover carried two antennas: a low-gain one with a wide beam for voice while driving, and an umbrella-shaped high-gain dish the astronauts aimed at Earth by hand for television when parked.
- **Not said: polarization.** The receiving rod has to line up with the wave's electric field. It is the real reason the CubeSat has two dipoles at right angles.
- **Not said: an isotropic antenna cannot be built.** It is a yardstick only.

SEC 5 · LEC 1 · VOCABULARY

Every word used, in plain English

48 entries. "Repeat" = you have met it before.

Telecommunications — sending information over a distance. Here: by radio between spacecraft and ground.

Signal / information — a quantity that changes in a pattern that means something: a voice, a picture, data.

Electromagnetic (EM) wave — a travelling ripple of electric and magnetic field. Radio and light are both this. *A ripple that needs no pond.*

Electric field / magnetic field — the region where a charge feels a push / where a magnet does. In a wave the two rise and fall together. *Electric field is a repeat.*

Perpendicular (⊥) — at right angles, like the corner of a page.

Sinusoidal / sinusoid / sine wave — the smooth, endlessly repeating up-and-down shape.

Variation / oscillating — changing / swinging back and forth regularly.

Voltage / difference in voltage — electrical 'height' between two points. It makes current flow. *Repeat, ion engines.*

Induce — to cause at a distance. The wave induces a voltage in the antenna.

Speed of light (c) — 299,792 km/s, about 3×10^8 m/s. The speed of every electromagnetic wave in vacuum.

Wavelength (λ, 'lambda') — distance from one crest to the next, in metres. *Repeat, UV in Section 2.*

Frequency (f) — how many waves pass per second.

Hertz (Hz) / kHz / MHz / GHz — one per second / thousand / million / billion per second.

Period — how long one wave takes: 1 ÷ frequency.

Constant — a value that never changes. c is one. *Repeat.*

Inversely related — when one goes up the other goes down.

Antenna — a metal conductor that turns electrical signals into radio waves and back. *A stick that bobs on the ripples.*

Transmitter / receiver — the electronics that drive the antenna / that read what it picks up.

Circuit — a loop of electronic parts that current flows through.

Dipole (half-wave dipole) — two rods end to end, half a wavelength long in total.

Dimension — a size: length, width, diameter.

Modulation — changing a carrier wave in step with a signal so the carrier carries it. *Writing on the ripples.*

Carrier signal — the fast, steady wave that does the travelling.

Information signal (baseband) — the slow signal you actually want to send.

Amplitude — the height of a wave.

Amplitude modulation (AM) — the carrier's height follows the signal.

Frequency modulation (FM) — the carrier's frequency follows the signal. Resists static better.

Phase modulation / PSK — shifting the carrier's timing to send bits. What spacecraft use.

Directivity — how strongly an antenna favors one direction.

Radiation pattern — the 3-D shape of where an antenna's power goes.

Isotropic antenna — an imaginary antenna that radiates equally in every direction. *Bare bulb.*

Homogeneous — the same everywhere.

Omnidirectional antenna — equal all around in one plane: a donut shape.

Directional antenna — sends most of its power one way. *Flashlight.*

Beam / beamwidth — the cone where the signal is strong / its angle in degrees.

Main lobe / side lobes — the big petal where most power goes / the small unwanted petals beside it.

3 dB / decibel (dB) — 3 dB down means half the power. Decibels count in multiplications: 10 dB = ×10.

Aperture / large aperture — the opening or collecting area of an antenna. Bigger aperture, narrower beam.

Gain / dBi — how many times stronger than isotropic in the best direction / the unit it is quoted in.

Parabolic antenna (dish) — a curved reflector that focuses radio waves into a beam.

High-gain antenna (HGA) — a narrow-beam dish for fast data. Has to be aimed.

Low-gain antenna (LGA) — a wide-beam antenna for slow, always-available contact.

Rooftop TV antenna (Yagi) — the fishbone-shaped directional antenna on houses.

Polarization — the direction the wave's electric field swings in.

CubeSat — a tiny satellite built from 10 cm cubes. *Who's Who.*

Cassini — NASA's Saturn orbiter, 2004–2017, with a 4 m dish. *Repeat, Who's Who.*

Apollo 17 / lunar rover — the last crewed Moon landing, 1972 / the car the astronauts drove. *Who's Who.*

Positioning / rotation = attitude — which way the spacecraft points. *Section 6.*

SEC 5 · LEC 1 · COPY INTO NOTES

Where each piece goes in the notebook

Routes A to D for this lecture. The concept cards (route E) are on the next two pages.

A · Equations & key numbers page (p3–4)

$c = \lambda \times f$	$c = 3\times10^8$ m/s (299,792 km/s)
	100 MHz → 3 m · 2.4 GHz → 12.5 cm
$\text{dipole} = \lambda \div 2$	100 MHz → 1.5 m · 100 Hz → 1,500 km
$\text{gain} \approx \frac{1}{2}(\pi D \div \lambda)^2$	4 m dish at 3.6 cm → ~50,000× (47 dBi)
$\text{beam} \approx 70 \cdot \lambda \div D$	→ 0.6°
Cassini	20 W → looks like 1 MW → 10^{-16} W at Earth

B · Section 5 map page (first sentence; more after the next lectures)

S5 · antennas: radio = EM wave, c = λ × f; rod = λ ÷ 2 so high f = small; modulation = signal on a fast carrier; gain = aim the power (dish), costs pointing

D · Question bank ("but how?" cards)

Q: How does a metal rod pick up a radio wave?
A: The wave's electric field pushes the rod's electrons to and fro → an alternating voltage at the wave's frequency.
Q: Why does the rod have to be about half a wavelength?
A: It resonates: it fits the wave, like a guitar string.
Q: How does a dish make a signal stronger with no extra power?
A: It takes the power going everywhere and aims it one way.
Q: How did 20 W reach Earth from Saturn?
A: A 4 m dish gave ~50,000× gain, and a 70 m dish listened.

C · One-line word bank (new lines)

p__ · electromagnetic wave · ripple of E + B field → C24
p__ · electromagnetic spectrum · radio … light … X-rays
p__ · wavelength (λ) · crest to crest, metres → C24
p__ · frequency (f) / hertz · waves per second → C24
p__ · period · seconds per wave = 1 ÷ f
p__ · speed of light (c) · 3×10^8 m/s; c = λ × f
p__ · sinusoid · the smooth repeating wave shape
p__ · amplitude · height of the wave
p__ · antenna · metal that turns signal ↔ radio wave
p__ · transmitter / receiver · sends / listens
p__ · dipole · rod λ ÷ 2 long; the basic antenna → C25
p__ · modulation · signal written onto a carrier → C25
p__ · carrier / information signal · fast wave / slow message
p__ · AM / FM · height follows signal / frequency follows it
p__ · phase modulation (PSK) · bits by shifting timing
p__ · isotropic · equal every direction; ideal only
p__ · omnidirectional · equal all around one plane
p__ · directional / beamwidth · one beam / its angle at half power
p__ · main lobe / side lobes · the beam / small leaks beside it
p__ · decibel (dB) · 3 dB = half or double; 10 dB = ×10
p__ · gain (dBi) · × stronger than isotropic → C26
p__ · HGA / LGA · narrow fast dish / wide slow rod → C26
p__ · polarization · direction the E field swings

SEC 5 · LEC 1 · CONCEPT CARDS

Route E: pages C24 and C25

Two notebook pages, two cards each. Copy line for line; the faint numbers are your ruled lines.

WORD BANK · CONCEPT CARDS · scene 11 — C24

ELECTROMAGNETIC WAVE = EM wave

= travelling ripple of electric +
magnetic field
WHY it needs no air or wire, so it is the
only way to carry information across
space
c = 299,792 km/s · radio, light, UV
all the same thing
≠ sound (air pressure, 343 m/s, no
vacuum)
→ c = λ × f · spectrum · antenna

WAVELENGTH + FREQUENCY = λ and f

= crest-to-crest distance / waves per
second
WHY speed is fixed, so the two are tied:
pick one and the other is set. λ sets
antenna size
100 MHz → 3 m · 2.4 GHz → 12.5 cm ·
8.4 GHz → 3.6 cm
≠ amplitude (the wave's height)
→ c = λ × f · dipole = λ ÷ 2

WORD BANK · CONCEPT CARDS · scene 11 — C25

HALF-WAVE DIPOLE = dipole antenna

= two rods end to end, λ ÷ 2 long in
total
WHY a rod that fits half a wave
resonates, so it gives the biggest
voltage for that frequency
FM 100 MHz → 1.5 m · Wi-Fi → 6 cm ·
voice 100 Hz → 1,500 km
≠ a dish (a reflector, many λ wide)
→ omnidirectional · monopole = λ ÷ 4

MODULATION = AM · FM · PSK

= changing a carrier wave so it carries
a signal
WHY voice-speed waves need 1,500 km
antennas. A fast carrier fits a small
antenna and its own channel
AM: 1 MHz carrier · FM: 100 MHz ± 75
kHz
≠ amplification (just makes it louder)
→ carrier · AM = height · FM = spacing

C24: 11 + 10 lines of 22 · C25: 10 + 10 lines of 22

Route E: page C26

The third card page for this lecture.

WORD BANK · CONCEPT CARDS · scene 11 — C26

ANTENNA GAIN = dBi

= strength in the best direction vs isotropic

WHY power spread over a whole sphere is wasted. Squeeze it into a beam and the far end hears more

\# dipole 1.6× · Cassini dish ~50,000× (47 dBi)

≠ added power (it only aims)

→ ≈ ½(πD ÷ λ)² · beamwidth ≈ 70λ ÷ D

HIGH-GAIN vs LOW-GAIN = HGA · LGA

= narrow-beam dish / wide-beam rod

WHY you want both: speed when you can aim, and a link that still works when you can't

\# Cassini HGA 4 m, 0.6° beam · 20 W looks like 1 MW

≠ transmitter power (same either way)

→ attitude control · safe mode · CubeSat

Why this lecture got two spreads

It is the longest lecture so far and it holds two separate ideas: how a wave carries a signal (waves, antenna size, modulation), and how you aim it (directivity, gain). Your rule is one spread per video unless it doesn't fit, and never smaller writing. This one doesn't fit.

If you are short on pages, copy spread 1 in full and add only the drawing from spread 2 under it. The gain idea is the one thing in spread 2 you can't skip.

You already know half of this. This is the physical layer under everything in networking: 2.4 GHz and 5 GHz Wi-Fi, antenna gain in dBi, why a directional antenna reaches farther. Use that. When Network+ resumes, this page is a head start on the wireless objectives.

C26: 10 + 10 lines of 22.

SEC 5 · LEC 1 · SAY IT OUT LOUD

Last page: prove it by talking

After the spreads and the 3-line summaries, with everything closed. Send me what you said and I will grade it like a friend would.

🗣 The friend's opening question

"How does a probe at Saturn talk to Earth? It can't have much power out there."

✅ Three sentences you should be able to say

- By radio, which is a ripple of electricity and magnetism that crosses empty space at the speed of light. An antenna is just a metal rod the ripple shakes.
- You can't send a voice or data directly, because slow signals would need an antenna thousands of kilometres long. So you write the signal onto a fast carrier wave. That is modulation.
- And it has very little power: Cassini had 20 watts. But a big dish squeezes it into a beam half a degree wide, which makes it tens of thousands of times stronger in our direction, as long as the dish points exactly at Earth.

❓ Follow-ups a friend might ask

- "Why does frequency matter?" (frequency times wavelength is always the speed of light, and the antenna has to be about half a wavelength)
- "AM and FM, what's the difference?" (AM changes the carrier's height, FM changes its spacing)
- "Does a dish add power?" (no, it only aims what is there, like a flashlight reflector)
- "What if the dish isn't pointing at Earth?" (you fall back to a low-gain antenna: wide beam, very slow)
- "Is that the same as Wi-Fi?" (same physics: 2.4 GHz is a 12.5 cm wave, and router antennas are a quarter of that)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Two more lectures in Section 5. Send them the same way.